JAYASURYA MARASANI

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AI Researcher

Jayasurya Marasani

I am an AI researcher with experience in machine learning, deep learning and reinforcement learning. My work spans computer vision, natural language processing, speech recognition and efficient AI for edge devices. I combine Python and C++ expertise with independent research, team collaboration and clear technical communication.

Education

2025 - Present | Milan, Italy

Politecnico di Milano

Master of Science in Computer Science and Engineering

2020 - 2024 | Amritapuri, Kerala

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science with Artificial Intelligence

CGPA: 9.09

Technical competencies

Languages

Python C++ Java MATLAB

Web development

HTML CSS JavaScript Django FastAPI React

Cloud & DevOps

Azure GCP Azure App Service Azure Functions Azure Blob Storage Cosmos DB GitHub Actions CI/CD

Machine learning & scientific computing

pandas NumPy scikit-learn TensorFlow Keras PyTorch SciPy matplotlib

Reinforcement learning

Gym Stable Baselines3

Robotics

ROS2 Gazebo OpenCV

Databases

SQL MongoDB Cosmos DB

Operating systems & documentation

Linux Windows LaTeX

Soft skills

Critical thinking Leadership Communication Project management

Experience

Professional roles and research collaborations.

Project Manager | Polimi Data Scientists (PMDS)

Feb 2026 - Present | Milan, Italy

Maxillary Canine Impaction

Lead team coordination, project execution and model development for the Maxillary Canine Impaction project.

Design an anatomy-prompted ROI attention network to predict canine impaction from panoramic radiographs.

Build a multi-task deep learning pipeline combining localization, classification and clinical risk scoring.

Apply landmark-guided feature fusion to improve interpretability and reduce ambiguity in 2D dental imaging.

BEA Smart Sleep Band

Contribute as a team member to AI-driven sleep monitoring on wearable hardware.

Develop a sleep phase detection model for local inference directly on the device.

Design for 256 KB RAM and a target model size of at most 50 KB.

Optimize memory use and inference efficiency for real-time on-device sleep phase detection.

Junior Developer - AI & Immersive Technologies | Ophir Studio

Oct 2025 - Feb 2026 | Milan, Italy

Prototyped and developed generative AI solutions for text, images, video and 3D across application domains.

Prototyped low-code MVPs for internal products.

Developed APIs and microservices to support scalable digital platforms.

Implemented CI/CD pipelines, containerization and cloud deployment workflows.

Collaborated with creative teams to integrate AI and immersive technologies into art and storytelling.

Contributed to research and innovation projects with universities and technology partners.

Undergraduate Research Assistant | Amrita Vishwa Vidyapeetham

Jan 2022 - Aug 2024 | Kerala, India

Led a five-person AI research team, contributing to more than 10 journal and conference publications in machine learning, deep learning, deep reinforcement learning and computer vision.

Led a three-person team and an international collaboration with Dr. Dugki Min at Konkuk University on DRL algorithms for image categorization. Achieved 97.57% accuracy on MNIST, outperforming ML/DL baselines across benchmarks.

Implemented a hybrid VGG16-XGBoost gaze estimation model, achieving 98.48% accuracy across datasets with varied environmental conditions.

Guided a three-person team developing a multi-agent DRL ensemble for autonomous galaxy classification, achieving 98.55% accuracy on Galaxy Zoo.

Developed a multimodal yoga pose recognition framework combining MoveNet, ResNet50 and attention fusion. Introduced the YFA-30 dataset and achieved 96% accuracy.

Projects

Applied AI systems, from efficient indoor localization to embedding model evaluation.

Depthwise Spatio-Temporal Heatmap Network for UWB Radar-Based Indoor Localization

Apr 2026 - Aug 2026

UWB Radar | TinyML | TensorFlow Lite Micro | Depthwise Separable CNN | INT8 Quantization | ESP32-S3

Developed a physics-guided TinyML system for device-free 2D localization of up to four people

using raw CIR measurements from six UWB radars. Combined static amplitude, motion, breathing-sensitive and inter-antenna phase features.

Designed a 22K-parameter fully convolutional network with temporal-sensor fusion, four depthwise separable residual blocks, and heatmap and person-count prediction heads.

Achieved F1 of 0.9430, precision of 0.9068, recall of 0.9821 and mean localization error of 0.2129 m on locked test windows using the fully quantized INT8 model.

Reduced the final TensorFlow Lite model to 42.2 KB with approximately 208.9 KB activation memory, within ESP32-S3 constraints of less than 800 KB model storage and 300 KB activation SRAM.

EmbedMatch: A Platform for Selecting the Best Embedding Model

Mar 2026 - Apr 2026

Azure | FastAPI | React | Azure Functions | Cosmos DB | Azure OpenAI | ONNX

Built an AI benchmarking platform to help users select embedding models for their own datasets.

Developed an asynchronous pipeline using Azure Blob Storage, Storage Queue, Azure Functions and Cosmos DB.

Integrated Azure OpenAI and ONNX-based local embedding models, including MiniLM and BGE.

Evaluated models using MRR, Recall@K, relevance score, latency and a weighted composite ranking.

Image Restoration

Django web application

Built a Django-based image restoration app with image upload, restoration and download features.

Research

Journal work and conference papers in reinforcement learning, computer vision, speech and language.

Journals

Publication statuses and journal metrics are reported as listed in my resume.

Leveraging Multi-Agent Deep Reinforcement Learning for Autonomous Galaxy Image Classification

The Egyptian Journal of Remote Sensing and Space Sciences | SCI-E | IF: 4.1

Mentor: Dr. Don S

Developed DRL agents using DQN, PPO, Recurrent PPO and A2C with customized policies and feature

extractors for galaxy image classification.

Combined agents through weighted majority voting for efficient classification.

Enhancing Yoga Pose Recognition Through a Multi-Modal Hybrid Approach with Attention Mechanism and Late Fusion on YFA-30

International Journal of Yoga | SCI-E | IF: 1.1

Mentor: Dr. Don S

Combined MoveNet’s efficient pose detection with ResNet’s feature extraction, attention mechanisms and late fusion for robust yoga pose classification.

Introduced YFA-30, a benchmark dataset designed for yoga pose recognition.

Adaptive Gaze Estimation Through Modified Deep Learning and XGBoost Fusion

SN Computer Science | Under review

Mentor: Dr. Don S

Developed a modified VGG16-XGBoost gaze estimation system for datasets with varied lighting, eye appearance and dimensionality.

Outperformed the compared deep learning and machine learning models, demonstrating potential for real-time gaze estimation.

Extended the work titled "Gaze Estimation Using VGG16 Architecture Through XGBoost Classifier".

Harnessing Deep Reinforcement Learning Algorithms for Image Categorization: A Multi Algorithm Approach

Engineering Applications of Artificial Intelligence | SCI-E | IF: 8.0

Mentor: Dr. Don S

Investigated multiple DRL algorithms for image categorization.

Demonstrated the potential of DRL for real-time computer vision challenges.

Conferences

10 June 2024

Speech Recognition Paradigms: A Comparative Evaluation of SpeechBrain, Whisper and Wav2Vec2 Models

9th International Conference for Convergence in Technology (I2CT)

10 June 2024

A Study of Landmark-Augmented and Landmark-Free Approaches: Yoga Pose Recognition

9th International Conference for Convergence in Technology (I2CT)

26 February 2024

An Empirical Analysis of Topic Categorization Using PaLM, GPT and BERT Models

2023 Innovations in Power and Advanced Computing Technologies (i-PACT)

26 February 2024

Analysis of Image Restoration Techniques on Lunar Surface Images

2023 Innovations in Power and Advanced Computing Technologies (i-PACT)

11 December 2023

Speaker Identification Using MFCC Feature Extraction: A Comparative Study Using GMM, CNN, RNN, KNN and Random Forest Classifier

2nd International Conference on Trends in Electrical, Electronics, and Computer Engineering (TEECCON)

23 November 2023

Examining Amazon Customer Reviews Using PySpark and AWS: A Data Lake Approach

14th International Conference on Computing Communication and Networking Technologies (ICCCNT)